

- 1. Design and implement Parallel Breadth First Search and Depth First Search based on existing algorithms using OpenMP. Use a Tree or an undirected graph for BFS and DFS .**
- 2. Write a program to implement Parallel Bubble Sort and Merge sort using OpenMP. Use existing algorithms and measure the performance of sequential and parallel algorithms.**
- 3. Implement Min, Max, Sum and Average operations using Parallel Reduction.**
- 4. Write a CUDA Program for : Addition of two large vectors CUDA C**
- 5. Implement HPC application for AI/ML domain.**
- 6. Mini Project: Implement Parallelization of Database Query optimization**
- 7. Linear regression by using Deep Neural network: Implement Boston housing price prediction problem by Linear regression using Deep Neural network. Use Boston House price prediction dataset.**
- 8. Classification using Deep neural network :- Multiclass classification using Deep Neural Networks: Example: Use the OCR letter recognition dataset <https://archive.ics.uci.edu/ml/datasets/letter+recognition>**

9. Classification using Deep neural network :- Binary classification using Deep Neural Networks Example: Classify movie reviews into positive" reviews and "negative" reviews, just based on the text content of the reviews. Use IMDB dataset

10. Convolutional neural network (CNN) :- Use MNIST Fashion Dataset and create a classifier to classify fashion clothing into categories.

11. Recurrent neural network (RNN) Use the Google stock prices dataset and design a time series analysis and prediction system using RNN.

12. Mini Project: Colorizing Old B&W Images: color old black and white images to colorful images